

Tribological properties of solid lubricants filled glass fiber reinforced polyamide 6 composites

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ABSTRACT

The main purpose of this paper is to further optimize the tribological properties of the glass fiber reinforced PA6 (GF/PA6, 15/85 by weight) for high performance friction materials using single or combinative solid lubricants such as Polytetrafluoroethylene (PTFE), ultra-high molecular weight polyethylene (UHMWPE) and the combination of both of them. Various polymer blends, where GF/PA6 acts as the polymer matrix and solid lubricants as the dispersed phase were prepared by injection molding. The tribological properties of these materials and the synergism as a result of the incorporation of both PTFE and UHMWPE were investigated. The results showed that, at a load of 40 N and a velocity of 200 rpm, PTFE was effective in improving the tribological capabilities of matrix material. On the contrary, UHMWPE was not conducive to maintain the structure integrity of GF/PA6 composite and harmful to the friction and wear properties. The combination of PTFE and UHMWPE showed synergism on further reducing the friction coefficient of the composites filled with either PTFE or UHMWPE only. Effects of load and velocity on tribological behavior were also discussed. To further understand the wear mechanism, the worn surfaces were examined by scanning electron microscopy.

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1. Introduction

Polyamides (PAs) are potential thermoplastic materials for various applications owing to their excellent comprehensive properties. It is also well known that fibers acting as a hard phase in PA can support the load preferentially apart from noticeably reducing interaction between the matrix and the counterface, so fibers are frequently incorporated into PA matrix to improve their friction and wear performances. Many studies have provided a better understanding of the effect of fibers' component, content and orientation together with the manufacturing process and the test conditions on the tribological properties. Li et al. [1], Kim et al. [2], and Shin et al. [3] observed that the glass fiber was effective in reducing the friction coefficient and the wear rate of the PA6-PU, PA12, and PA66 respectively. And Cartledge and Baillie [4] found that the wear resistance of GF/PA6 composite was significantly influenced by the thermal history of the composite. A slow cooling given the GF/PA6 a better tribological behavior than that of a fast cooling did. Carbon fiber has been also widely used to reinforcing PA composite. Li and Xia [5] reported that carbon fiber filled PA6 had superior tribological characteristics to unfilled PA6. The optimum wear reduction was obtained when the content of carbon fiber is

20 vol%. Nie et al. [6] indicated that surface treatment of carbon fiber could effectively improve the interfacial adhesion between carbon fiber and PA6 matrix. Thus the wear resistance was significantly improved. While Cai and Wang [7] revealed that the friction and wear for CF/PA66 slightly in increase with the increase in applied pressure. The friction was in decrease while the wear was in increase with the increase in velocity. AF is a kind of high-tech synthetic fiber. Researches on the AF reinforced composites were widely studied in recent years. Xu et al. [8] investigated the tribological properties of AF reinforced PA1010, and found that the addition of AF decreased the friction coefficient of PA1010 and increased its wear resistance. Some researchers also revealed the machining of fiber reinforced composites. Gaitonde et al. [9,10] have carried out experimental studies on machinability in PA6 and PA66 reinforced with 30% glass fiber (PA66 GF30) with K10 carbide tool. They found that the requirement of machining force, cutting power and specific cutting pressure is higher in PA66 GF30 polyamides machining as compared with PA6 polyamides in a given condition.

Fiber reinforced polymeric composites (FRPCs) form an important class of tribo-materials due to their good combination of high specific strength and specific modulus. They are widely used in a variety of structures in most varied fields, from automobile industry to biomechanics. Fuchs et al. [11] reported that fiber-reinforced polymer composite technologies offered a way of light-weighting the vehicle. Davim and Marques found that PEEK filled with 30%

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carbon fiber showed better performance than pure PEEK when used as an acetabular socket of a total hip prosthesis [12]; They also reported that the counterface roughness and unidirectional sliding velocity revealed a great effect in the friction behavior of both materials while contact stress led to an almost negligible effect [13]; At the mean time, they compared the effect of the reinforcement carbon fiber or glass fiber in the tribological behavior of the PEEK at long dry sliding and a constant $p_v = 2$ MPa/m/s, the obtained results showed that PEEK-CF30 presented the better tribological behavior than that of PEEK-GF30's [14].

To make the fiber reinforced polymeric composites tribologically even more outstanding, the solid lubricants are additionally needed. Tang et al. [15] investigated the effect of nano-MoS₂ on the tribological properties of aramid fiber reinforced PA6, they found that with the addition of nano-MoS₂, worn debris multi-layer laminar film was found on the wear scar, MoS₂ transferred prior to PA6 and the friction coefficient and wear rate decreased. Wang et al. [16] reported a wear and friction reduction in carbon fiber reinforced PA1010 being filled with MoS₂. They demonstrated that the improved properties were related to the increase in bond strength between the transfer film and counterface. As solid lubricant, UHMWPE containing straight molecular chain of $-CH_2-CH_2-$ is another obvious candidate. Hashmi et al. [17] reported that UHMWPE efficiently improved the tribological property of cotton fiber reinforced polyester composites. It is well known that both PTFE and graphite can make the positive contribution to the formation of transfer films, hence greatly reduce the friction coefficient and wear rate. So they have been more widely used to enhance the tribological properties of fiber reinforced polymers. Jia et al. [18] reported that the incorporation of carbon fiber and solid lubricants (PTFE or graphite) contributed to decrease the friction coefficient and wear rate of PI to various extents under both dry and water lubricated sliding. Moreover, PI and its composites recorded further decreased friction coefficients under water-lubricated sliding than under dry sliding. Shiao and Wang [19] found that the addition of PTFE and fibers decreased the delamination phenomena of epoxy and reduced the wearing. And an improvement in tribological behavior obtained when graphite added to fibers reinforced epoxy composite due to graphite has a high thermal conductance coefficient that made the heat dissipate quickly from the contact surface. Zhang et al. [20] showed the combinative of carbon fiber and PTFE or graphite enhanced the tribological properties of epoxy: short carbon fibers were the main reinforcement, which contributed to the improved wear resistance significantly; PTFE or graphite reduced the friction coefficient. Bahadur and Polineni [21] added PTFE to glass fabric-reinforced and CuO-filled nylon composites, and found that PTFE was very effective in reducing wear due to the role of transfer film. However, it should be noted that the solid lubricant may not be able to improve the friction and wear performance of all composites, or to improve the friction and wear performance under all test conditions. Larsen and co-workers [22] showed that PTFE had positive effect on improving the wear resistance of carbon/aramid weave reinforced epoxy, but had negative influence on the friction coefficient at the same time. On the contrary, PTFE reduced the friction coefficient and improved the wear resistance of glass fiber reinforced PA6 [23]. While Bijwe et al. [24] found that PTFE enhanced the wear resistance as well as reduced the friction coefficient of CF reinforced PA12 in both adhesive and fretting wear modes, but showed exactly opposite trends in abrasive wear mode. In summary, the solid lubricants can either enhance or degrade the tribological properties of polymer composites. The final results closely relate to the constituent of composites, the content of solid lubricants, and the test conditions and so on.

Our previous study [25] has demonstrated that PTFE and UHMWPE improved the tribological properties of PA6, but there

is few research about tribological properties of solid lubricants modified glass fiber reinforced PA6. In this paper, PTFE, UHMWPE and their combination were detected as fillers of GF/PA6 composite. One of the objectives of this work is to study the effects of filler content, applied load and velocity on the tribobehavior of the composites. The other is to discuss the synergism of fillers on the improvement of tribological performance of the composites.

2. Experimental details

2.1. Materials

The continuous glass fiber reinforced PA6 selected in this study (GF/PA6, 15/85 by weight) was purchased from the Shenzhen Changtai Plastic (Shenzhen, China). The diameter of the glass fiber ranged from 13 to 24 μm . It was dried under a vacuum at 100 °C for 6 h to remove excess moisture. PTFE was supplied by Tangshan Ri Feng Chemical (Tangshan, China). The size of the particle was less than 1 μm . UHMWPE, average molecular weight $M_n = 3$ –6 million, was supplied by Shanxi Zhong Ke Tian Gang Technology Development (Lvliang, China).

2.2. Specimens preparation

The compounds of different solid lubricants with GF/PA6 were achieved by using twin-screw extruder ($\varnothing = 35.5$ mm and $L/D = 41$) with a screw frequency of 360 rpm and a feeding frequency of 20 Hz under processing temperatures of 210, 220, 230, 240, 240, 240, 240 and 240 °C in nine zones of the extruder barrel. The extrudate was continuously cooled by water and pelletized. The pellets were dried in oven at 100 °C for 6 h again ready for injection molding. Standard test samples were injection molded, the injection pressure was 45 MPa, and the temperatures in four zones were 250, 250, 250, and 250 °C and in the nozzle 230 °C. The molded specimens were visually inspected for air bubbles and those with defects were discarded.

2.3. Friction and wear analysis

Friction and wear test were performed according to Chinese Standard – GB 3960-83 [26] on an UMT-3 (Universal Macro Materials Tester) friction and wear tester. Columnar structure chromium steel with a $\varnothing 9.5$ ball (HRC 62) was used as the counterface. The dimension of the sample designed by the size of machine fixture's, not according to the standard description size, is $25 \times 25 \times 15$ mm³. The specimens were cleaned ultrasonically with acetone and thoroughly dried before testing. Dry sliding under ambient condition with two loads (40 N and 100 N) and two velocities (200 rpm and 1500 rpm) were performed in a trace of 10 mm and duration of 120 min. The friction coefficient was recorded and calculated by the ratio between the tangential force and normal load. The average values of friction coefficient in the test range were used as the friction coefficient of samples. The width of the wear tracks were measured with a KH-7700 digital microscope to an accuracy of 0.001 μm . Then the wear rate was calculated using the following equation:

$$\text{wear rate} = \frac{B}{Ld} \left[\frac{\pi r^2}{180} \sin^{-1} \left(\frac{b}{2r} \right) - \frac{b}{2} \sqrt{r^2 - \frac{b^2}{4}} \right] [\text{mm}^3/\text{Nm}] \quad (1)$$

where B is the trace of friction (10 mm), r is the semi diameter of the chromium steel ball (mm), b the width of the wear trace (mm), L the load (N) and d the sliding distance (m). Each of the tribological data shown in Table 1 was the average value of at least three specimens with identical experiment parameters.

2.4. Analysis of worn surfaces

The worn surfaces were inspected by Scanning Electron Microscope (SEM, JSM-6363LV). In order to increase the resolution for the SEM observation, all the tested samples were coated with a thin gold layer to endow them with electric conductivity.

3. Results and discussions

3.1. The effect of added solid lubricants

3.1.1. PTFE filled GF/PA6 composite

PTFE possesses an extraordinary characteristic of very low friction coefficient while its wear resistance is poor. Therefore, it is commonly used as solid lubricant to other more robust polymer matrix, presumably with the intention of reducing the friction coefficient and lowering the wear rate.

Fig. 1 shows the variations of friction coefficients and wear rate of GF/PA6 composite with different PTFE contents tested at 40 N load and 200 rpm speed. As seen from the figure, the friction coefficient and wear rate decrease with the increase in PTFE content in all test range. As can be seen from the results given in Table 1, glass fiber can reduce the friction coefficient and improve the wear resistance of PA6, and the friction and wear properties can be further improved by the addition of PTFE. The composite with 20 wt%PTFE is of best wear resistance, and the friction coefficient and wear rate decrease by 30% and 29% compared with GF/PA6, respectively. In a word, the addition of PTFE is likely to improve the wear resistance of GF/PA6 composite. The above observation is in accordance with what has been reported elsewhere [23]. The low friction coefficient seems to be related to the reduction of fiber fracture, which results in less plowing friction compared with GF/PA6 without PTFE, because there are fewer rigid fibers ends in the contact surface. This phenomenon was also found in long woven fiber reinforced PEEK [27]. The wear seems to be closely related to the fiber fracture and abrasion as well as transfer film formation on the counterface. It is well known that PTFE can form a third-body transfer film on the counterface easily. During the process of sliding, PTFE transfers prior to PA6 due to the low surface energy of PTFE and the easy sliding of chains in banded crystalline PTFE. As the rubbing goes on, breakages of the molecule chain of PTFE take place at C–C or C–F bonds, thus the F-ions generated from the breakage of C–F bonds will react with the metal counterface, and then the transfer film will chemically bond to the counterface, forming strong adherence. As a result, the wear rate of the composite is reduced since the transfer film is much more difficult to destroy [28].

3.1.2. UHMWPE filled GF/PA6 composite

The tribological properties of GF/PA6 composite as the function of UHMWPE loading at 40 N and 200 rpm are shown in Fig. 2. It can be seen that, the addition of UHMWPE to GF/PA6 leads to the increase in friction coefficient and wear rate. Looking back to Table 1, both the friction coefficient and wear rate of GF/PA6 composites with UHMWPE are higher than that of pure PA6 except for that containing 5 wt%UHMWPE. Our previous paper [25] showed that UHMWPE was effective to improve the tribological behaviors of PA6. Thus the positive effect of UHMWPE on friction and wear seen for UHMWPE/PA6 is not found for UHMWPE/GF/PA6. Larsen et al. [22] reported a similar result showing the positive effect of fillers on tribological properties seen for EP composite was not found for CA/EP composite. They concluded that the friction and wear behavior were largely controlled by the fibers and not the particles. Shiao and Wang [19] reported three different mechanisms of fiber reinforced polymer: (1) The matrix and fibers were worn thin by the counterface. (2) Debonding between the interface of fibers

and matrix. (3) Finally, the fibers broke, detached and the matrix were removed. Maybe in this case, the lubricant effect factor of solid lubricant is not as significant and the fiber fracture is most likely to be responsible for the higher friction coefficient and wear rate. In other words, the reasons that why UHMWPE have negative impact on tribological properties of UHMWPE filled GF/PA6 composites can be listed as follows: because of the compatibility of UHMWPE and PA6 is not so good, or the interface adhesion between PA6 and UHMWPE is weak, consequently the structure integrity of the composite will be destroyed with the addition of UHMWPE which leads to more fibers fracture, and the high-level fiber debris at the sliding interface are likely to disrupt or scuff the soft transfer film owing to its abrasive nature, as a result the wear and friction performances are deteriorated. Fig. 2 also shows that the friction coefficient has a decreased trend when the UHMWPE loading over 10 wt%, which is explained by that there are much enough UHMWPE debris in the sliding interface functioned as lubricants.

3.1.3. Combinative of PTFE and UHMWPE filled GF/PA6 composite

Possible synergy effects of PTFE and UHMWPE were examined. In this paper, the total loading of solid lubricants was fixed at 15 wt%. The effect of various proportions of two solid lubricants on the friction and the wear behavior of GF/PA6 composites are presented in Fig. 3. Compare with the composite containing 15 wt%PTFE, the combination of PTFE and UHMWPE have a beneficial impact on the friction behavior but no positive impact on the wear behavior. In this figure, the lowest friction coefficient is demonstrated by the composition with 12 wt%PTFE/3 wt%UHMWPE; the wear rates of all the composites incorporated with combinative lubricants are slightly higher than contained 15 wt%PTFE alone. Meanwhile combinative lubricants exhibit more excellent lubricity than UHMWPE. The wear and friction properties of the combinative lubricants filled composites improved with the increase in PTFE proportion.

Fig. 4 reveals how the solid lubricants effected on coefficient with sliding time. It is clear that the variation of the friction coefficient of the PA6/GF composite demonstrates three friction stages, a short initial increase stage, followed by a decrease stage, and then a slight increase stage. This is analogous to the results reported by Kim et al. [2] of the glass fiber reinforced PA12. The composite with UHMWPE indicates an initial rapid increase stage followed by a decrease stage, and then a steady stage, and compare to GF/PA6, both the peak value and the duration of the running-in stage is increased. Unlike those two materials, the composite with PTFE suggests two friction stages, a slight increase stage followed by a steady stage, and the peak value is lower than that of GF/PA6's. The friction of the composite with both PTFE and UHMWPE exhibits similar behavior as the friction curve of composite with PTFE. But the curve is smoother and the erratic phenomenon at the end of the other three curves is not obtained in this case, the peak value is further lowered and the time to reach the steady state friction level is earlier as well. This is believed to be due to the synergism of PTFE and UHMWPE. As well documented in the literatures [29–31], PTFE effectively reduced the peak value and the duration of the running-in stage due to the quick formation of a stable transfer film. Thus in this case the wear surface is smoother and the loose fibers at the wear interface maybe reduced which is benefit for UHMWPE as lubricant, then the erratic phenomenon disappears. In conclusion, the composite filled with PTFE and UHMWPE achieves the smoothest curve and the lowest value of the friction coefficient.

3.2. The influence of applied load and velocity

Fig. 5 is a bar chart represented the variation of friction coefficient and the wear rate with load and velocity. As one can see from

Table 1

Wear results of the composites under various loads and velocities.

Compositions				Friction coefficient			Wear rate (10^{-5} mm ³ /Nm)		
PA6 (weight)	Glass fiber (weight)	PTFE (weight)	UHMWPE (weight)	40 N 200 rpm	100 N 200 rpm	40 N 1500 rpm	40 N 200 rpm	100 N 200 rpm	40 N 1500 rpm
100	0	0	0	0.2111			2.769		
85	15	0	0	0.2063	0.1942	0.1255	1.645	3.528	10.943
85	15	5	0	0.1904			1.605		
85	15	8	0	0.1794			1.582		
85	15	10	0	0.1735			1.483		
85	15	12	0	0.1693			1.347		
85	15	15	0	0.1614	0.1197	0.0830	1.248	2.73	0.329
85	15	20	0	0.1446			1.170		
85	15	0	5	0.2100			2.174		
85	15	0	8	0.2213			2.917		
85	15	0	10	0.2392			3.174		
85	15	0	12	0.2353			3.847		
85	15	0	15	0.2209	0.1899	0.0985	4.156	4.192	0.602
85	15	0	20	0.2152			5.997		
85	15	3	12	0.2254			2.505		
85	15	5	10	0.1906	0.1869	0.0975	1.853	3.688	0.515
85	15	7	8	0.1751	0.1740	0.0883	1.682	3.440	0.510
85	15	10	5	0.1531	0.1635	0.0961	1.583	3.308	0.449
85	15	12	3	0.1515			1.485		

Fig. 5a that the friction coefficient decreases with the increase of load or velocity for both GF/PA6 and its composites. The result of GF/PA6 is different from GF reinforced PA6/PU [1] and PA12 [2] for which the friction coefficient increased with load and decreased with velocity. But it is consistent with the studies by Xu et al. [8] and Unal et al. [32] who reported the friction coefficient of fiber reinforced polymer composites decreased with the increase of load. Therefore, the different effects of load on friction coefficient should be attributed to the characteristic of composite. It is well known that the friction of polymers resulting from adhesion equal the product of the real contact area and the shear strength of the polymer [33]. With the increasing load, the contact temperature between the polymer composite and the counterface will generally increase due to the frictional heat, which results in two competitive effects on the friction coefficient. On the one hand, the shear strength of the matrix material decreases which is benefit to reduce the friction coefficient. On the other hand, the matrix soften at higher temperature which results in an increase in the real contact area and then the higher friction coefficient obtained. Therefore the final friction coefficient of the polymer will be determined by these two competitive aspects. In the present case, the friction coefficient of PA6 and its composites with load follows the equation of $\mu = K \times N^{(n-1)}$, where K is constant, N is the load and n is also a constant, its value $2/3 < n < 1$. The equation reveals that the coefficient of friction decreases with the load increase which is documented in literatures [24,27]. The effect of sliding velocity on friction coefficient of materials mainly attributes to the generation of friction heat. The accumulation of friction heat promotes a glass-to-viscoelastic-to-viscous flow state transition. When the velocity increases over a critical value, the composite is in viscous flow state, and the viscosity and shear strength of the surface material decrease. Accordingly, the friction coefficient decreases. In addition, Fig. 5a also shows that at a higher load or velocity, a reduction in friction coefficient is detected in all of the solid lubricants filled composites with a constant amount of 15 wt%.

The wear rates for differing load and velocity are represented in the form of bar diagram in Fig. 5b. It becomes clear that the wear rate of the GF/PA6 matrix composite without solid lubricants increase with the load or velocity. Li and Xia reported a similar result from carbon fiber reinforced PA6 showing an increased friction and wear under higher load and velocity [5]. It can be attributed to the friction-induced heat effects. At higher load and velocity, the

surface temperature will increase and the glass fiber may easily be chipped off from of the matrix due to the weakened bonding between them; the fibers also tend to be broken and crumbled, and the fiber debris will be deposited on the counterface acting as large abrasives, thereby the wear rate will be elevated. With respect to the wear rate of the composites with solid lubricants, the influences of load or velocity are different from each other. It is seen in Fig. 5b that the wear rates increase with the increase of load and the decrease of velocity. As mentioned previously, with the increase of load, more small fragments are deposited onto the counterface due to the higher colliding dynamics and higher contact temperature. The fragments shall rub away the transfer film during the repetitive sliding; they may also overshadow the lubricant effect of solid lubricants and consequently result in higher wear rates. This is similar to the tribological result of PA12/CF/PTFE composite showing an increase in wear rate with increase in load [24]. The dependence of wear rate on velocity can be interpreted as follows: at higher velocity, the fibers may be trapped by the lubricant film on the worn surface then serious fiber fracture avoided. Those appear to facilitate the formation of fine, uniform and continuous transfer film on the counterface thereby result in better wear resistance. It is obvious that the wear rate of GF/PA6 composite reduced more than an order of magnitude via being modified with solid lubricants, in other words the solid lubricants have enlarged the range of applied velocity for GF/PA6 composite.

3.3. SEM analysis of the worn surface

The SEM micrographs of the worn surfaces of the composite are represented in Fig. 6. Fig. 6a–d shows the worn surface of GF/PA6 composites with or without 15 wt% solid lubricants on the constant condition of applied load 40 N and velocity 200 rpm. In the case of composite GF/PA6 without solid lubricants (see Fig. 6a), ploughing traces are observed in the worn surface. The cavities are formed due to the removal of PA6. It is suggested that adhesion and abrasive wear appear to be the main wear mechanism. Cavities seen in Fig. 6a are not observed when PTFE added into the composite as seen from Fig. 6b and glass fiber exposure is hardly detected which conformed to the significantly improved tribological behavior. This observation is consistent with what has been reported elsewhere [18]. It can be explained by the role of PTFE covering the fibers in total: during sliding, when the PTFE's back-transfer

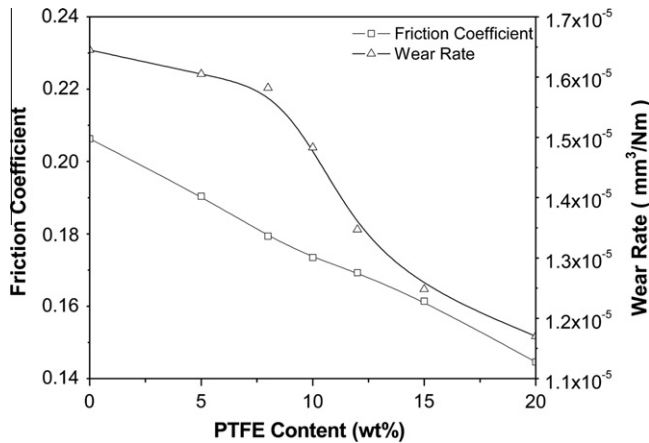


Fig. 1. Variation of friction coefficient and wear rate of GF/PA6 composite with PTFE at 40 N and 200 rpm.

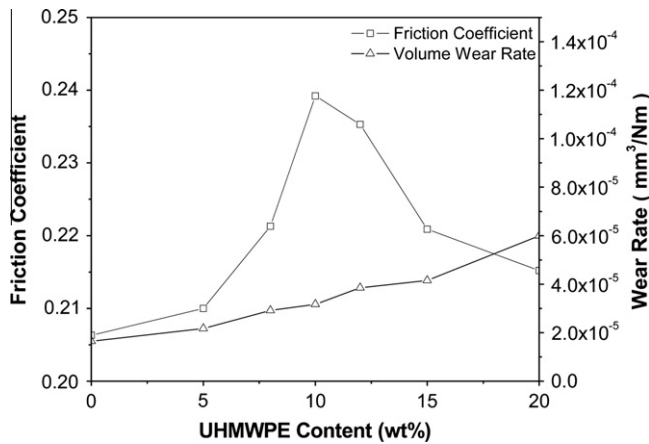


Fig. 2. Variation of friction coefficient and wear rate of GF/PA6 composite with UHMWPE at 40 N and 200 rpm.

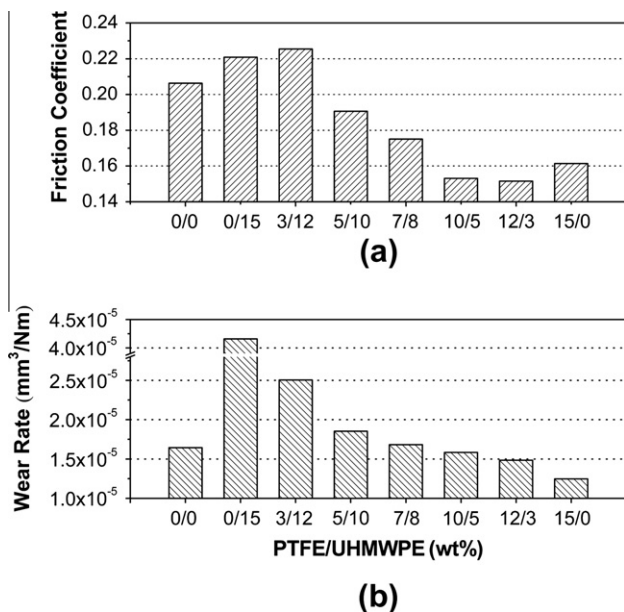


Fig. 3. Variation of friction coefficient and wear rate of GF/PA6 composite with both PTFE and UHMWPE at 40 N and 200 rpm. (a) Friction coefficient and (b) wear rate.

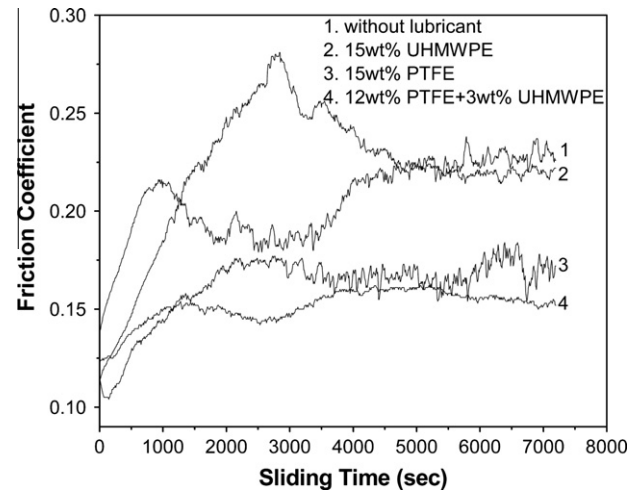


Fig. 4. Variation of coefficient of different composites with sliding time at 40 N and 200 rpm. (1) GF/PA6 without solid lubricant; (2) with 15 wt%UHMWPE; (3) with 15 wt%PTFE; and (4) with 12 wt%PTFE and 3 wt%UHMWPE.

from the counterface to the polymer surface occurs, the PTFE lubricant film formed, which will decrease the contact and adhesion between polymer composite and counterface, and finally result in a reduce in friction and wear. The stripes on the worn surface indicate that the viscous flow have occurred by reason that friction induced heat. Microcracks are also seen perpendicular to the sliding direction due to slight fatigue wear. It is reasonable to conclude that viscous flow, material transferring and fatigue wear may be the important mechanism. The worn surface of composite with UHMWPE is shown in Fig. 6c. The interface between UHMWPE and PA6 is obviously seen, the interface debonding is also detected due to the stress concentration which correspond to the poor tribological properties. Once the fibers are removed from the matrix, they will rub the matrix in the following sliding process. Even more, when the fiber fragments collide with other fibers, they can cause further breakages of fibers. Those may result in a high wear rate. In short, the interface debonding, fiber fractures, fiber crushes and cavities left after peeling off of fibers seem to be the main characters of this micrograph (Fig. 6c). Fig. 6d shows the worn surface

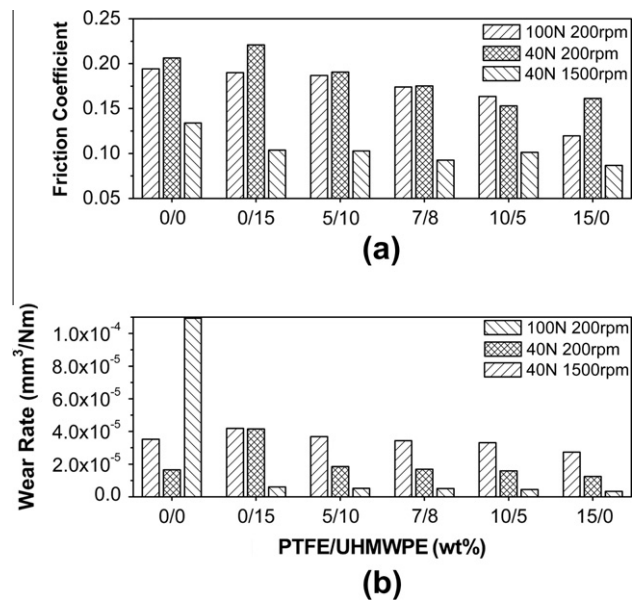


Fig. 5. Effect of load and velocity on the friction coefficient and wear rate of GF/PA6 composites with and without 15 wt% solid lubricants. (a) Friction coefficient and (b) wear rate.

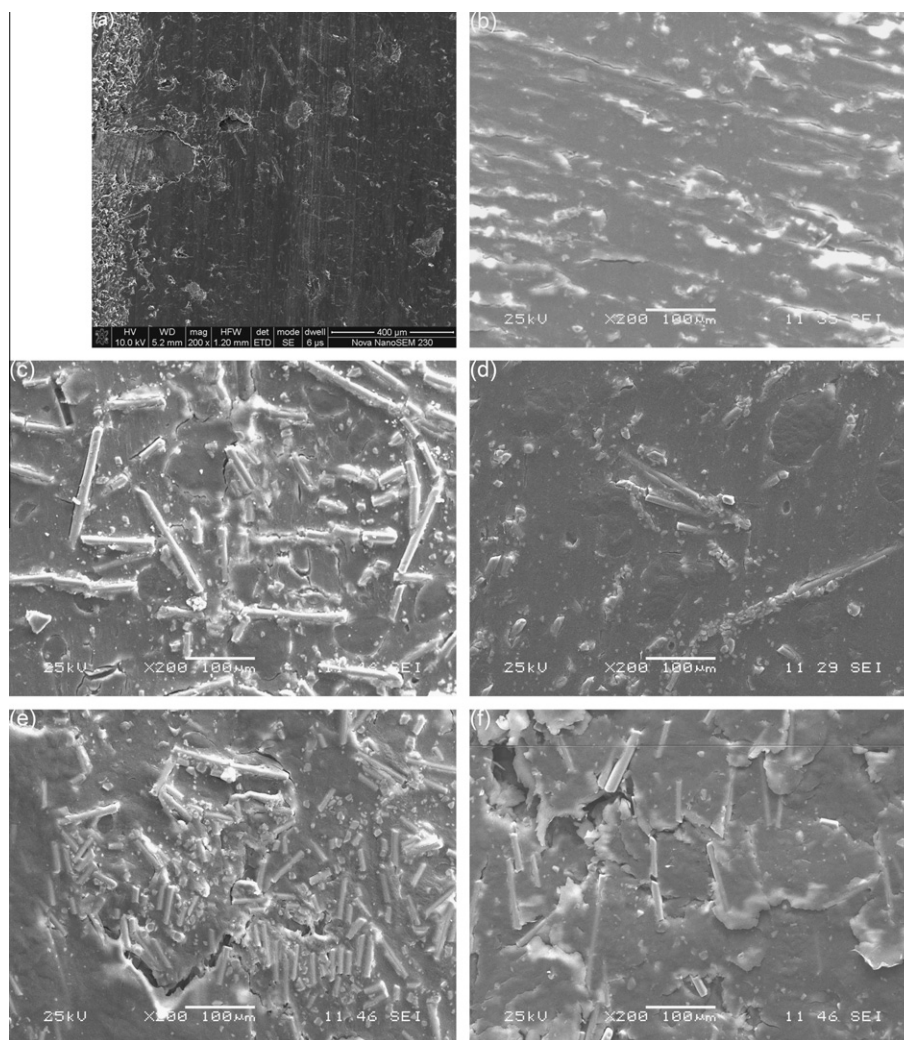


Fig. 6. SEM morphologies of the worn surface of different composites. (a) GF/PA6 without solid lubricant; (b) with 15 wt%PTFE; (c, e and f) with 15 wt%UHMWPE; and (d) with 5 wt%PTFE and 10 wt%UHMWPE. Image a–d at 40 N, 200 rpm; image e at 100 N, 200 rpm; and image f at 40 N, 1500 rpm.

of composite with combinative solid lubricants, compare with Fig. 6c, fiber fractures alleviated despite some tendencies to fragment which may be responsible for a relative lower wear rate than UHMWPE modified composite.

The worn surface of 15 wt%UHMWPE modified composite at higher load and velocity are shown in Fig. 6e and f. Fig. 6e illustrates in comparison to lower load 40 N (Fig. 6c) more destroyed worn surface: much more fiber fractures are observed and the cracks become more obvious which result in a relative high wear rate. While with sliding velocity of 1500 rpm as can be seen in Fig. 6f, back-transfer of the material to the polymer surface is found, the wear debris tend to accumulate near the glass fiber and act as lubricant film, then prevent the glass fiber from further fracture, such a film may be account for a reduce in wear rate.

4. Conclusions

The friction and wear of GF/PA6 and its composites were studied. Based on the results in the present study, the major conclusions can be drawn as follows:

1. Under 40 N and 200 rpm, with the addition of PTFE into GF/PA6, the friction coefficient and wear rate of GF/PA6 composites decreased substantially. The optimum wear resistance property is obtained at the PTFE content of 20 wt%, in this case, there is a 30% reduction of the friction coefficient and a 29% reduction of the wear rate compared with GF/PA6. It is found that the addition of UHMWPE cannot improve the wear resistance of GF/PA6. However, the combined solid lubricants shows synergism effect on reducing the friction coefficient. The lowest friction coefficient is obtained in composite filled with combinative solid lubricants (12 wt%PTFE and 3 wt%UHMWPE), while it is higher when only one solid lubricant (either 15 wt%PTFE or 15 wt%UHMWPE) is filled.
2. At the given test conditions, the friction coefficient of GF/PA6 shows a decrease tendency with the increase of load and sliding velocity and the wear rate increases with the increase of load and velocity. For solid lubricants filled GF/PA6 composite, the friction coefficient decrease with the increase of load and velocity, but the wear rate shows an increase tendency with the increase of load and a decrease tendency with the increase of velocity.
3. The main wear mechanisms of GF/PA6 are adhesion and abrasive wear. When incorporated with PTFE, the removal of PA6 reduced; the viscous flow, material transfer and fatigue wear are the important mechanisms. When added with UHMWPE, the fiber fracture became the main mechanism due to the poor interfacial bonding.

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